

A Robotic Hand: Controlled With Vision Based Hand Gesture Recognition System

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Abstract—Human-robot interaction (HRI) has become an important topic in today's robotic world, especially in assistive robotics. Now pen, mouse, keyboard etc. are not sufficient due to the limitations of these devices for taking command. Human now wants direct use of hands as an input device for getting more natural human-robot interactions (HRI). Vision based hand recognition system has come with a solution for these type of demands of human. In this system, human need not to operate any kind of devices, instead a camera is used for recording the movements of hands and these recorded data are used as input for performing any kind of gesture control. The elderly people, patients and disabled can be benefitted from these kind of gesture control. Besides it can be used in extreme environments where human can't go directly. Here, a vision based hand gesture system for controlling robotic hand has been developed. The main intension here, is to implement a vision based gesture recognition system that recognizes data gathered from hand movements through a camera.

Keywords—Human Robot Interaction, Hand Gesture Recognition, Robotic Hand.

I. INTRODUCTION

With the development of the modern technology in our society, we are very much dependent on the computers for our daily life work. These will necessitate the development of new and updated forms of human-computer interaction (HCI), with user-friendly and natural interfaces. From a text-based command to a graphical interface with mouse and keyboard inputs, the computer user interface (UI) has evolved. However, they are unnatural and inconvenient. These input devices like mouse, pen, keyboard etc. have some limitations too as they need direct human hand interactions. Besides, human now wants more easy way for giving inputs to any device. They don't want any direct physical touch. As a result, vision based hand recognition system has become very popular in recent years. For providing more natural human-robot interactions (HRI) hand can be used as a data input device. Vision has the ability to carry a large amount of data in a nonintrusive and low-cost manner. As a result, it's a particularly appealing sensing method for recognizing hand motions [1]. The shape and orientation of hand movements are acknowledged in a hand gesture recognition system to task the system into performing some work. In a vision-based hand gesture recognition system, a camera detects the movement of the hand. This input video is then segmented into a number of frames and then each frame is taken into account. Unwanted data are removed and highlighting of necessary components

are done through some form of filtering. Numerous gestures can be made by a person at a time through which it can provide large number of data input. Hand gesture recognition is not only confined in controlling robotic hand, it has a large field of applications. It can be used in car controlling, playing and pausing of a music audio or video, bomb disposal, chemical handling, radio-active material handling, sign language, medical system and in many other difficult conditions which are inaccessible to humans like, emergency hostile conditions or some remote areas. Physically operating the devices or appearing physically in certain working settings could be extremely risky for human operators. In such cases this type of gesture control robotic hand can perform beyond imagination [2].

II. RELATED WORKS

The hand gesture recognition technology can be thought to be a very recent technology, but the first research was held around 40 years ago in 1980 by the name of "Put-That-There". Though the research was at the initial state, it opened a large area for future research work. Researchers have done many researches in this field of 'Gesture Recognition'. They have made this field easy to understand and added many features to improve gesture recognition. At the beginning researchers used data-gloves to capture hand shapes and hand movements. R.Raja Prabhu and R.Sreevidya focused on handling industrial job through gesture controlled robotic arm [3]. Dao Tuan Minh, Tran VanHuy, Nguyen Phan Kien, Tran Anh Vu used flex sensors to control a prosthetic hand through servo motors [4]. Suzette Antoniette Dela Cruz, Paolo Creas Carando, Neil John Lorena, Juan Paolo Zabat, and Alexander C. Abad developed a system where they controlled a robotic arm through flex sensor for handling hazardous chemicals. Here they showed the usefulness of gesture controlled robot in hazardous conditions [5]. But vision-based gesture recognition has emerged in this field and made it more acceptable in the recent years. A video-based sign language recognition system was developed by Starner and Pentland which was an impressive achievement in this field [6]. C. W. Ng and S. Ranganath developed a real-time recognition system for 14 motions based on vision [7]. K. Abe, H. Saito, and S. Ozawa created a system that identifies hand motions by detecting the bending of the hand's five fingers [8]. This work was quite impressive and earned others attractions. Hasanuzzaman used skin color segmentation and multiple-feature based template-matching approaches to create a real-time hand gesture detection system. The three largest skin-like regions from the input images were segmented

using a skin color segmentation technique and they were compared for feature-based template matching [9]. Nielson suggested a real-time vision system that extracts the hand gesture from the entire image using a rapid segmentation procedure. It detects the hand posture from the temporal sequence of segmented hands. 26 hand postures were recognized by the system and achieved a 90% recognition average rate [10]. Elena et al. used a matching procedure to recognize hand movements, in which the Hausdroff distance was used to compare the segmented hand with all the saved gestures in the system's memory [11]. All the gestures, their edge maps, their distance transforms and morphologic information are stored in the system's memory. Triesch and Malsburg described a method that is extremely robust in real-world settings [12]. Color characteristics derived from a set of training photos and an elastic-graph-match technique were used to improve recognition efficiency. A gesture recognition system which is able to detect different hand gestures under a cluttered background condition was represented by Y. Cui [13]. But the system has relatively slow segmentation speed and is not user independent. Hidden Markov Models (HMM) have some inherent qualities that make them particularly appealing for dynamic gesture identification in a variety of illumination settings. T. Starner used HMM to learn repeatable patterns of human gestures to develop a gesture recognition system to recognize sign language. The recognition rate of the pattern was 92% [14]. A hand-signed digit recognition system was implemented by Jonathan Alon, Quan Yuan, Vassilis Athitsos and Stan Sclaroff. Their algorithm can recognize movements using a relatively simple hand detection technique that provides multiple candidates. The system works without causing any disturbance for users wearing short-sleeved shirts in the presence of a busy background, several moving objects, multiple skin-colored image areas [15]. This was the most advanced work done by any researcher.

III. METHODOLOGY

The project methodology can be divided into two parts, one is method for capturing hand gesture and the other method is to control the robotic hand. For capturing the hand gestures, OpenCV and MATLAB plays an important role. In this project, I have used OpenCV for capturing hand gestures. OpenCV, is an open source computer vision programming function library planned for creating applications dependent on ongoing computer vision advancements. The implementation of the robotic hand is nothing but a mechanical work. Proper wiring and servo control have been done here for the controlling of the robotic hand.

A. Workflow Diagram

In this project a robotic hand is controlled through vision based hand gesture recognition system. Here a live video camera is used to capture the hand gesture. If the frame contains any gesture then it is cropped out and applied further processing. Then the gesture produces a particular binary output. This binary output is different for each gesture. The output signal is then sent to Arduino. Finally, instructions in Arduino are executed for the matched gesture, that actuate the servo motor to control the robotic finger.

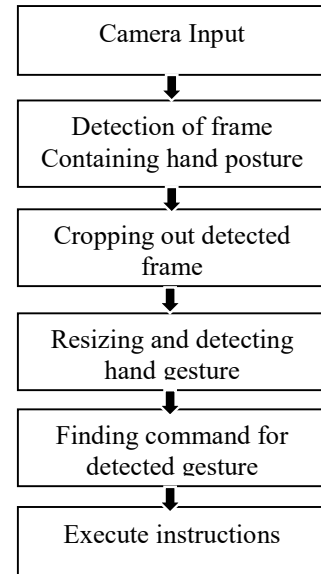


Fig. 1. Flow Chart for Detecting Hand Gesture

B. Detecting Fingertips

For hand gesture recognition, the first thing to do is the detection of the fingers. For this, a frame containing a hand is taken and a binary mask is created in order to compute the hand contour. Next step is to find all contours in the mask. After the detection of contour, the computation of convex hull as well as the convexity defect regions of the hand contour is done for detecting finger tips. The edge points of the hull locates at the fingertips. Any deviation of the object from the hull can be considered as convexity defect and the defect points are located in the valley position between two fingers. The angle of the valley is determined by cosine theorem. If the angle is less than 90 degree then we treat it as a finger [16]. Figure 2 shows the detection of by cosine theorem. The angle γ can be derived from:

$$\gamma = \cos^{-1} \left(\frac{a^2 + b^2 - c^2}{2ab} \right)$$

C. Output Generation and Servo Control

The frame that has a gesture contains extra parts such as blank spaces, background objects etc. These excess unused parts must be separated for a better result. As a result, the obtained part is cropped to remove the hand gesture. The cropping is done with three steps. At first the cropped frame is converted into binary which is black and white using global thresholding. Secondly, object of interest is extracted from the frame.

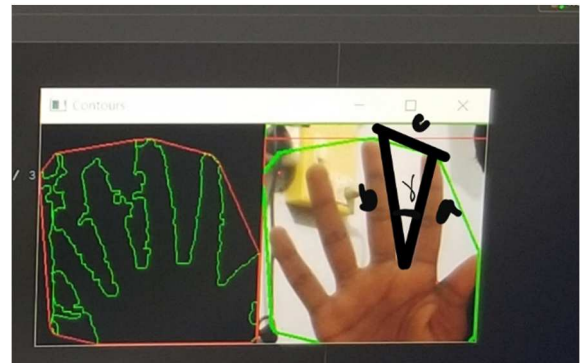


Fig. 2. Detecting Fingers by Cosine Theorem

Thirdly, the wrist, arm, and other portions of the hand that are not involved in gesture display are removed. Because the extra components in the image frame are of varying lengths and can give unwanted results. After cropping, cropped gesture gives a particular output for each gesture. This output is produced according to the finger positions. Five outputs based on finger positions have been used in this project.

TABLE I. OUTPUT BASED ON GESTURES

Gesture	Pinky	Ring	Middle	Index	Thumb
Zero	0	0	0	0	0
One	0	0	0	1	0
Two	0	0	1	1	0
Three	0	1	1	1	0
Four	1	1	1	1	0
Five	1	1	1	1	1

0 = Close

1 = Open

The outputs generated from the hand gestures are sent to Arduino UNO. A 5V battery is used to power the Arduino. Five servo motors are connected to and controlled with this Arduino. Each motor is also powered through the 5V battery. And each motor is connected to a robotic hand finger through a nylon string. After receiving the outputs the Arduino actuates the servo motors according to the outputs.

TABLE II. LIST OF MOTORS CONNECTED TO EACH FINGER

Fingers	Servo motors
Pinky	Motor 1
Ring	Motor 2
Middle	Motor 3
Index	Motor 4
Thumb	Motor 5

If the generated output is 00010 then the Arduino will actuate Motor 4 which is connected to the index finger. If the output changes to 00110 then Motor 4 and Motor 5 both will be actuated at the same time. Again if the output changes to 11111 then all the motors will be actuated at the same time with the fingers attached to each motor.

IV. RESULT

A. Robotic Hand

In this project, the robotic hand used is a human like hand shown in Figure 3. There are five fingers and three joints in each finger. Every finger is connected to a servo motor through nylon string. Motors control the fingers according to the gestures. An Arduino is used to store data and actuate the servo motors through the command from gesture recognition.



Fig. 3. Robotic Hand

B. Hand Gesture Recognition

In this project the hand gestures recorded from a live video camera are used to control a robotic hand. Here five commands for five hand gestures are used. The algorithm enables the robotic hand to detect that five possible hand gestures and performs as commanded. The identified command can be used as a control unit for the robotic hand and therefore it can perform certain tasks as programmed. Some commands based on finger positions used in this project are given below:

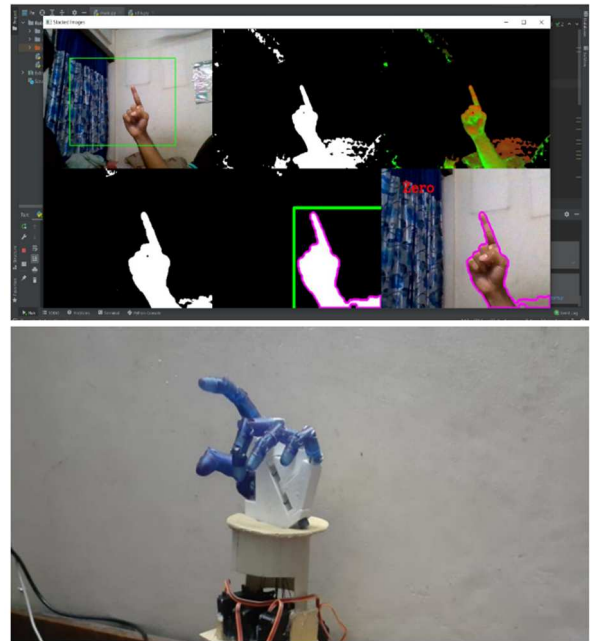


Fig. 4. Gesture Command(one) for Robotic Hand

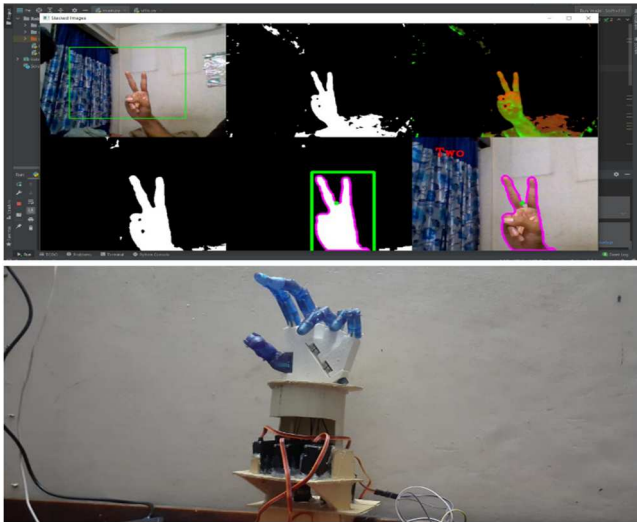


Fig. 5. Gesture Command(two) for Robotic Hand

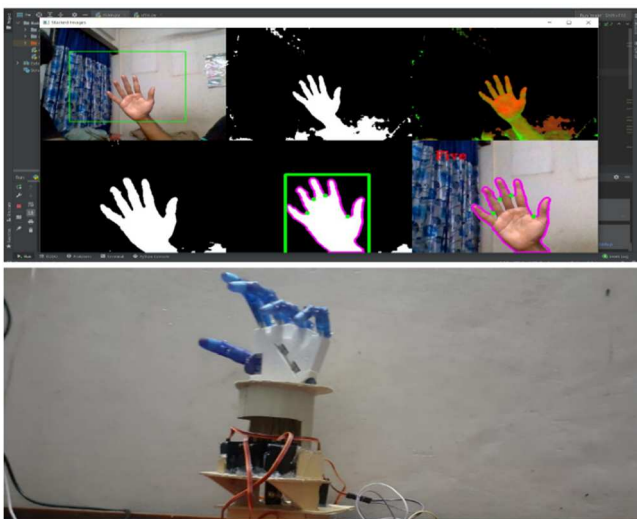


Fig. 6. Gesture Command(five) for Robotic Hand

In this project hand gesture is captured through live video camera. But this causes a problem when operating the same program in different light conditions. In hand gesture recognition, RGB (Red, green, Blue) image is converted to HSV (Hue, Saturation, Value) image. Then thresholding is done to reduce background noise. But if the light of the room changes, we need to change the HSV value manually for getting a good result which creates difficulties during operations. The hand gestures from different hands sometimes cause problems because of different skin color. Besides the servo motors used here can not actuate the fingers properly as the fingers are quite hard to move and this is due to the mechanical structure of the robotic hand.

Though the performance of vision based gesture interaction is prone to be influenced by illumination changes, speed of the system satisfies real time requirements. The illumination problem can be solved by setting the HSV value manually which gives it the flexibility to use it under any lighting condition

V. CONCLUSION

In the present digitalized world, technologies have expanded drastically, with PCs being developed to the

extent where they can help people in complex tasks. However, input devices appear to cause a noteworthy limitation. Vision based hand gesture recognition has a great advantage in this case. In recent years, it has become a popular area for research work. In this project a total implementation of a robotic hand controlled with vision based gesture recognition has been given. A limited (five) number of hand gestures have been considered here. Though the number of gestures can be extended by adding different algorithm, the computation will be increased. OpenCV library is preferred here as it is applicable for real time and execution is faster. Its application area lies to sign language through medical science. It may be used on a variety of robotic systems with a set of gesture commands appropriate to that system.

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